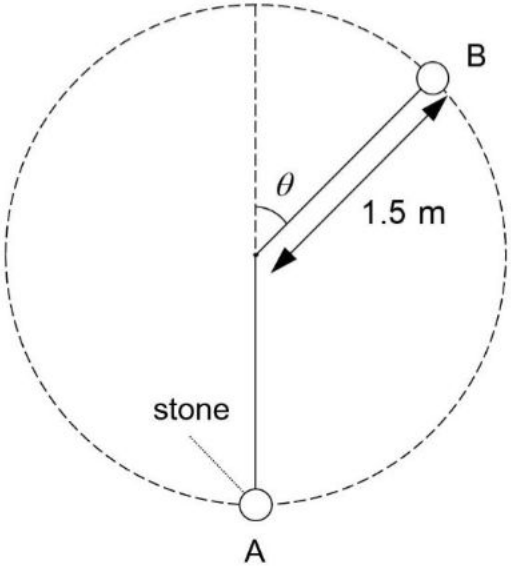
7 A stone attached to a light inextensible string at position A is given an initial push to the right. It subsequently moves in a vertical circle of radius 1.5 m and the string just slackens at position B when it makes an angle of $\theta$ to the vertical.




Given that the velocity of the stone at position B is 2.2 m s$^{-1}$, what is $\theta$?

**A** 9° **B** 19° **C** 71° **D** 81°

Ans: C

When the string just slackens, the tension in the string is zero. The component of the weight acting in the radial direction provides the centripetal force.

$$mg\cos\theta = \frac{mv_B^2}{r}$$

$$g\cos\theta = \frac{v_B^2}{r}$$

$$\theta = \cos^{-1}\left(\frac{v_B^2}{rg}\right)$$

$$= \cos^{-1}\left[\frac{2.2^2}{(1.5)(9.81)}\right]$$

$$= 71°$$

